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Article

Determinants of FPI in Developed and Developing Countries

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Abstract

The pattern of capital inflows in developed and developing economies are different because of dissimilar economic and political structures. From the point of view of host country, especially the developing countries, portfolio flows are considered to play a pivotal role in bridging the saving investment gap and providing foreign exchange to finance current account deficit. While the investors of developed country invest in portfolios of different countries to diversify the risk and earn more returns, foreign portfolio investors generally go for short-term investment to reap the benefits of good economic conditions and they tend to withdraw their investments during the period of recession. This article identifies the determinants of foreign portfolio investment (FPI) in developed and developing economies. Though the movement of capital among different countries is researched in depth by existing literature, the present study adds to literature by identifying the institutional factor involving freedom index. The institutional factors aid in identifying the determinants of FPI among select developed and developing countries. This study seeks to answer, where the funds of foreign portfolio investors are headed. And also the reasons of attractiveness for FPI among different sets of countries. The sample of the study is limited to a set of 19 developed and developing counties for the period of 10 years (2004–2013). We study the determinants of FPI for a group of developed and developing countries using fixed and random effects. Additionally, we use panel generalized method of moments (GMM) suggested by Arellano and Bond (1991, *The Review of Economic Studies*, *58*(2), 277–297). This methodology is suitable to remove the problem of endogeneity which static model is not able to capture. The results of model also incorporates persistence effect considering lagged value of dependent variable. The study empirically tests the various factors that determine the inflows of FPI and analyses their performance during different stages of the economic cycle in the last 10 years. Implicitly, in case of developed countries, it was observed that interest rate differential, trade openness, host country stock market performance and US stock market returns are significant trendsetter, while in developing countries, freedom index, interest rate differential, host country stock market performance, trade openness, US stock market returns and crisis period (2006–2008) significantly influence the inflow of FPIs. Dynamic model supports that as a group of 19 countries, portfolio investments are significantly influenced by interest rate differentials, freedom index, US stock market and host country stock market returns.

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Keywords

Foreign capital inflows, determinants, foreign portfolio investment, static and dynamic panel

Introduction

The growing removal of restrictions on trading of international financial assets has led to surge in the flow of capital across the world. The two forces, globalization and liberalization are the roots of extraordinary foreign capital inflows and have become the engines of economic development. The process of globalization increases competitiveness in environment, leading to higher efficiency in the system. Implicitly, all economies are initiating major reforms so as to make their countries more vibrant and strategically competitive. The economies are making conscious efforts to attract foreign financial capital to achieve the objective of higher economic development. Most developing countries are achieving the twin goals of macroeconomic stabilization and structural adjustments to reduce the country's borrowing fiscal imbalances. The main focus of these policies are formulated to balance external debt and trade deficits coupled by the process of fiscal, monetary and industrial reforms being set with great zeal. Globalization leads to integration of home economy with the world by forming global financial relationship. The dimensions of financial globalization include abolishing credit controls, deregulation of interest rates and finally allowing foreign direct investment (FDI) and foreign institutional investment (FII) in multiple sectors of economy in the form of capital flows. Foreign capital inflows have significant role in the developmental process of all countries. Developed countries require foreign capital inflows for sustainable development, substantially developing countries need it for higher economic growth and augmenting domestic investment critical for improving the quality of life of majority of people. It is a channel through which developed as well as developing countries have access to foreign capital, which basically comes in two forms: (a) FDI and (b) foreign portfolio investment (FPI). FDI and FPI are similar as they both originate from foreign investors, but fundamental difference exists which is the degree of control. FDI investors exercise controlling position in domestic firms and actively getting involved in management, while FPI investors are passive investors not actively involved in day-to-day operations and strategic plans of domestic companies. Foreign investment has emerged as the engine of economic growth in a globalized world and is considered as one of the important areas in the study of international business. Research in this area has been found to be very extensive, but few deficiencies are observed in literature. First, the most of the research on foreign investment are focused on FDI, whereas the topic of foreign indirect investment or foreign portfolio investment (FPI) has received less attention (Li & Filer, 2007). Second limitation highlights that the effect of governance environment has been left out while considering the determinants of FPI. Later on, Wu, Li and Selover (2012) has taken into account government environment to determine the inflow of foreign portfolio investment. The prior literature suggests that economic development level, financial markets, geographical or cultural factors are important for foreign capital inflow. On the one hand, these are certainly the factors that determine the inflow of foreign capital, and on the other hand, literature is silent about institutional factors and freedom index. How does freedom index play a role? If it really plays a role, how and why does it play a role? In this study, we try to fill the gap taking into consideration freedom index (as a measure of institutional factors) of the countries as one of the important determinant in performance of foreign capital inflows.

The study deals with foreign portfolio investment which refers to investment made by residents of a country in financial assets and production process of another country, including FII, American depository receipts (ADR) and global depository receipts (GDR). The effects of foreign investment varies from country to country. It can affect the balance of payment, productivity, financial markets and stock market

performance. Foreign portfolio investment is a short-term investment, mostly invested in financial markets of the host countries. Here in this article, we focus on two questions: first is that how relevant macroeconomic variables (including institutional/freedom index) are in affecting FPI inflows and second one is that how important common and global shocks have been for the inflow of foreign capital.

Global Trends in FPI (Foreign Portfolio Investment) Inflows

The decade (1998–2008) immediately preceding the global financial crisis observed an exponential surge in worldwide capital inflows and outflows, especially in emerging markets. The favourable trend towards emerging market is because of encouraging demographic characteristics, potential growth and scope of diversification and economic policies, which lead to inflow of high foreign portfolio investment in BRICS countries (Garg & Dua, 2014; Ghosh & Herwadkar, 2009; Srinivasan & Kalaivani, 2015). Global quarterly foreign portfolio index[1] highlights the trend and performance of foreign portfolio investment. Due to global slowdown in 2008, trends of foreign capital inflows drastically decreased all over the world, but capital tend to increase in later years after recovery in the global slowdown due to Asian Financial Crisis. Despite the global slowdown both types of foreign capital inflows (FDI and FPI) showed deterioration trends. Foreign portfolio investment (FPI) declined during the third quarter of 2008 because of global financial crisis, and again due to double-dip recession during the second quarter of 2011, it comes to its inferior level.

Theoretical Background

Foreign investment includes two categories: direct investment and portfolio investment. In direct investment, investors invest in a firm with an intention to have the right to participate in the management of the firm, whereas in case of portfolio investment, investors purchase the securities of firm to earn short-term returns foregoing the interest of ownership. The present study is considering the determinants of FPI and theoretical base for the same is listed:

1. Portfolio balanced framework: Under this model, foreign investors exploit all the possibilities of arbitrage process present in home and host countries. The empirical model of the present study is motivated by factors suggested in theoretical model of balanced framework. This framework analyses the effects of domestic and global factors on capital flows. Balanced portfolio framework studied pull and push factors that attract foreign portfolio investments. The pull factor represents the domestic country-specific investment risk and returns which attracts foreign investment, and push factors represents global liquidity and other factors that push investments to emerging economies (Mody, Taylor, & Kim, 2001).
2. International finance theory: According to this theory, FPI are foreseeable outcome of investors, those who wish to invest across the countries. Investment in different portfolio leads to diversification of risk and higher returns can be achieved. There are several studies made which documented

   (i) the benefits of diversification on portfolio,
   (ii) international funds which bridges the gap of saving and investment across the countries, and
   (iii) reviewed the benefits of capital flows from host country perspective (Dell'Ariccia et al., 2008; Obstfeld, 2009).

3. Capital allocation theory: To follow diversification, foreign investors can invest in either emerging markets or in financial markets of industrialized countries. As per Buckberg (1996), foreign investors monitor two-step process in determining the capital allotment. In the first step, they identify the amount of investment to be allocated; then in the next phase, they allocate the capital in each emerging market on the basis of earnings. This simply indicates that emerging market with high inflow of capital points out the high potential of a country in terms of investment. In other words, this theoretical model proposed to take the industrial and economic growth of a set of emerging markets to identify the potential among them. This simply directs to incorporate stock market returns to catch FIIs. In this way, emerging stock market returns are also included in the proposed empirical model.

In an attempt to fill the gaps in literature, the article is organized as follows: The previous section covered an overview of global trends of foreign portfolio investment and theoretical framework. The next section presents the literature review on determinants of foreign portfolio investment in developed and developing countries. The third section describes research design, including time period, data sources and the econometric methodology employed. The fourth section presents the empirical results and findings of the study. The fifth section concludes the article and the final section covers managerial and policy implications.

Literature Review

Foreign capital inflows have been studied extensively and its importance is widely established in empirical literature. Theories suggest determinants of potential capital inflows are likely to fall in three categories: country-level, industry-level and firm-level determinants. We focus on country-level determinants, including economic and political factors that measure the trends of capital inflows. A snapshot of literature review is given in Figure 1.

Inflow of FPI has been gaining importance as it increases the financial strength of the host country. Contrary to that, capital flight may cause vulnerability in host country. To increase the financial strength of the country, continuous inflow of FPI is required, which is supported by various lightening factors. Researchers stressed on pull and push factors that affect the trends of portfolio investments. Many studies enlist global factors such as interest rates, growth of industrialized countries and many more that have pushed inflow of FPI to developing countries (Byrne & Fiess, 2011; Calvo, Leiderman, & Reinhart, 1993, 1996; Felices & Orskaug, 2008; Kim, 2000; Mody et al., 2001; World Bank Report, 1997).

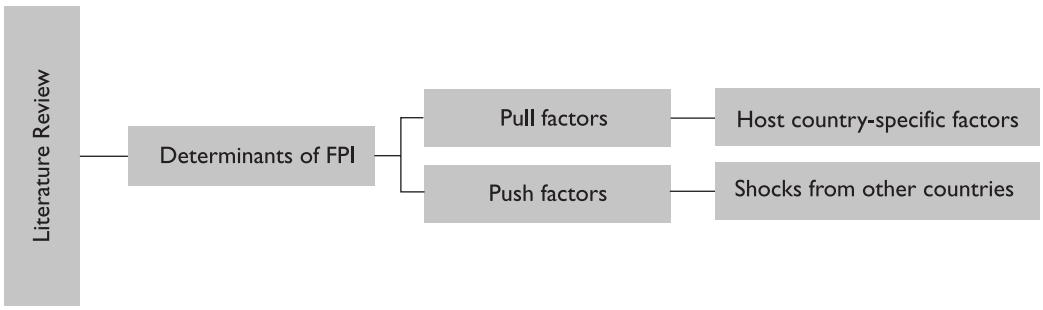

Figure 1. Snapshot of Literature Review

Source: Authors.

Another set of studies consider domestic as well as external factor influencing FPI in developed and developing countries (Chuhan, Claessens, & Mamingi, 1998; De Vita & Kyaw, 2007).

Many studies support foreign capital inflows and their benefits to economy through the process of diversification (Dell'Ariccia et al., 2008; Grubel, 1968; Harvey, 1991; Obstfeld, 2009). The constant removal of restrictions on trading of international financial assets has led to surge of capital across the world during last two decades, and the investment is geared fast towards the developing countries, including BRICS, ASEAN and developing Asia (Garg & Dua, 2014). Developing countries, especially BRICS, attracted considerable amount of foreign capital in terms of direct and portfolio investment wherein China is the highest recipient. Holtbrügge and Kreppel (2012), Morck, Yeung and Zhao (2008) and Mostafa and Mahmood (2015) studied BRICS and G7 countries in terms of development, productivity, markets, consumptions, saving, investments, human capital, innovations, potential and challenges, and predicted that BRICS has the prospects to challenge G7 countries in the coming decades in terms of inflows of foreign capital. The detailed chronological analysis of literature is presented in Table 1.

Research Design

Objective

The objective is to investigate the influence of institutional context of a country in determining the FPI inflows. There is a long tradition of research in international business based on determinants of FPI inflows. The literature has identified multiple factors ranging from macroeconomic to political variables. We propose to identify various economic, political and institutional factors ranging from host country stock market returns to US stock market returns, interest rate differentials to credit worthiness of the country, institutional variables to exchange rates, affecting FPI inflows.

Rationale of the Study

This article makes an important contribution to the existing literature by adding an empirical analysis based on determinants in select developed and developing countries using static and dynamic panel data approach. Most of the studies earlier done in the area of foreign portfolio investment are based on time series data, specific to a country, but the present study tried to incorporate 19 countries in the form of panel data. The study will consider many macroeconomic variables, including pull and push factors such as interest rate differential, freedom index, stock market performance of host country along with US stock market returns with special consideration to crisis period that determine the performance of portfolio investment in host countries among developed and developing countries. Without doubt, many researches have analysed this propaganda, but there are little documented evidences of the existing literature on factors influencing FPI in developed and developing countries. Therefore, the proposed article aims to identify the factors that affects the inflow of FPI in developed and developing countries.

Sample Countries and Period of Data

To analyse the determinants of FPI, 19 countries, including 11 developed and 8 developing countries (enlisted as top 22 recipients of foreign capital inflows prescribed by World Bank report 2012 and 2013), have been selected and presented in Table 2.

Table 1. Chronological Analysis of Literature Review on Determinants of FPI

S. No.	Author(s)	Title of Study	Sample Data	No. of Sample Country	Variables	Methodology/ Tools Adopted for Data Analysis	Findings and Conclusions
1	Agarwal (1997)	Foreign portfolio investment in some developing countries: A study of determinants and macroeconomic impact	1986–1993	6	Inflation rate, real exchange rate, index of economic activity, domestic capital market	Multiple regression analysis	Determines the foreign portfolio investment in four Asian countries where inflation is having negative significant impact and real exchange rate, index of economic freedom, domestic capital market are having positive significant impact on foreign capital inflows.
2	Montiel and Reinhart (1999)	Do capital controls and macroeconomic policies influence the volume and composition of capital flows? Evidence from the 1990s	1990–1996	15	Portfolio flows, short-term flows, FDI, as the percentage of GDP	Static panel data: Using fixed and random effects	Proposed that the direction and magnitude of capital flows between emerging and industrial economies depends upon the relative attractiveness of investing funds in host countries and three broad categories, pull factors, push factors and changes in degree of financial integration highly affects the inflow of foreign capital.
3	Chakrabarti (2001)	FII flows to India: Nature and Causes	1993–2001	1	BSE return, MSCI world index return, S&P-500, return volatility, exchange rate.	OLS, Chow Breakpoint Test	FII flows are correlated with returns of Indian stock exchange. US returns, MSCI returns, country-risk rating for India is affecting FII inflows. However, domestic and international variables also affect FII inflows.
4	Rai and Bhanumurthy (2004)	Determinants of FII in India: The role of return, risk and inflation.	1994–2002	1	FII inflows, domestic and foreign inflows, ex ante risk factor and crises dummy	TGARCH	The study results into the dependence of foreign institutional investment (FII) on stock market returns, inflation rate and ex-ante risk. Stabilising stock market volatility and minimising the ex-ante risk would help to attract more FIIs, which has positive impact on the real economy.

5	Portes and Rey (2005)	The determinants of cross-border equity flows	1989–1996	14	Purchase and sale of equity, distance, telephone volume, market capitalization, number of bank branches, degree of insider trading, presence of financial market, stock returns	Fixed effects	The study considers the importance of gravity model in explaining the determinants of foreign equity capital inflows. It is found that market size, efficiency in transactions, technology and distance are the most important determinants of equity flows.
6	Boyer and Zheng (2009)	Investor flows and stock market returns	1952–2004	1	Stock market returns and high frequency data of stock market	VAR and regression analysis	Using VAR and regression analysis, it was found that the quarterly contemporaneous relations between stock market returns and flows are positive and significant for mutual funds and foreign investment during the period of study.
7	Lin, Lee and Chiu (2009)	Structural changes in foreign investors' trading behaviour and the corresponding impact on Taiwan's stock market	1995–2005	1	Stock market and FIIs	ARMA, ARCH, GARCH-ARJI Model	The expected trading behaviour of foreign investors increases stock return volatility, provided if there is gradual increase in foreign trading limits.
8	Bohl and Brzeszczyński (2009)	Institutional investors and stock returns volatility: Empirical evidence from a natural experiment	1994–2003	1	Stock market returns FII inflow	Markov Switching GARCH analysis	Institutional investors stabilize the stock market prices.
9	Byrne and Fiess (2011)	International capital flows to emerging and developing countries: National and global determinants	1993–2009	78	Real commodity prices, short-run US interest rate, long-run US interest rate, volatility, real GDP	Pooled panel regression model	It was found that for long-run US interest rates to be a crucial determinant, real commodity prices have relatively small but somehow played an important role in capital flows to EMEs.

(Table 1 continued)

(Table 1 continued)

S. No.	Author(s)	Title of Study	Sample Data	No. of Sample Country	Variables	Methodology/ Tools Adopted for Data Analysis	Findings and Conclusions
10	Ghosh Qureshi, Kim, and Zalduendo (2014)	Surges	1980–2009	56	Net capital flow, real US interest rate, S&P 500 return index, real GDP interest rate, trade openness, reserve to GDP, capital account openness index, institutional quality index, exchange rate, commodity price index, 3 month treasury bill	Probit, 2SLS and country fixed effect	Identify episodes of capital inflow surges and find a variety of factors to be important in increasing the likelihood of a surge to EMEs, including lower US interest rates, greater global risk appetite and a particular EME's own attractiveness as an investment destination.
11	Fratzscher (2012)	Capital flows, push versus pull factors and the global financial crisis	2005–2010	50	Growth, foreign exchange reserve, current account balance, inflation, interest rate, short-term debt, bank credit growth, financial institution, trade openness, market capitalization and different crises period.	Pooled panel regression and step wise regression model.	The findings of the study reflects that push factors in the form of shocks to liquidity, risk, micro-economic conditions and policies in advanced economies, particularly in USA, have exerted a significant effect on capital flows to EMEs and advanced economies, although these effects are larger during financial crisis period of 2007–2008 on inflows of foreign capital. This study highlights the importance of pull factors and concludes that countries with sound economic policies and improvement in institutional setting resist to the risk of external shocks.

12	Forbes and Warnock (2012)	Capital flow waves: Surges, stops, flight and retrenchment. *Journal of International Economics*	1980–2009	50	Volatility index, trade openness, bank claim among countries, GDP growth rate, interest rate, growth, trade, GDP per capita	Multiple regression with sensitivity analysis.	Global factors, especially the global risk, predict the sudden stops in capital flows by foreign investors and retrenchment by domestic investors. Decrease in global risk surge capital inflows in country, and concludes that global interest rate is not affecting the surge and stops in foreign capital inflows.
13	Bhasin and Khandelwal (2013)	FII in India: Determinants and impact of crises	1994–2011	1	US SP 500, MSCI index, 3 month LIBOR, BSE index, lagged BSE returns, volatility of monthly returns, exchange rates, call money rate, IIP, WPI, lagged Value of dependent variables	Factor analysis for data reduction then multiple regression analysis is done.	Return on MSCI emerging market index, past values of FIIs and growth rate of economy (IIP) are considered as the most important determinant meaning that both domestic and global factors affects the inflow of foreign capital.
14	Dua and Garg (2013)	Foreign portfolio investment flows to India: Determinants and analysis	1995–2011	1	Domestic stock market performance, exchange rate, currency risk, country risk, stock return risk, risk diversification, global diversification, global liquidity, interest rate differential, MSCI returns	ARDL	The results suggested that India may be able to attract portfolio flows by maintaining strong domestic growth, lower exchange rate volatility and making domestic financial market less vulnerable to shocks.

(Table 1 continued)

(Table 1 continued)

S. No.	Author(s)	Title of Study	Sample Data	No. of Sample Country	Variables	Methodology/ Tools Adopted for Data Analysis	Findings and Conclusions
15	French and Vishwakarma (2013)	Volatility and foreign equity flows: Evidence from the Philippines	2003–2006	1	FII, exchange rate and stock market returns	SVARX-GARCH	Foreign equity investors are trend chaser and equity flows are auto correlated. It was found that unexpected increase in foreign equity flows to a country increases the conditional volatility of stock market significantly over the upcoming next two weeks of trading. Also the unexpected shocks to foreign equity flows increases the conditional variance of exchange rate over next two or three weeks.
16	Ahmed and Zlate (2014)	Capital flows to emerging market economies: A brave new world?	2002–2013	12	Growth differential, US treasury 10 years, crisis dummy, global risk aversion	Static panel data using fixed effect	The study considers the aspect which is related to growth and interest rate differentials between EMEs and advanced economies, along with this global risk appetite is also found as statistically and economically important determinants. US policy framework is one among the several important factors influencing capital inflows.

17	Srinivasan and Kalaiyani (2015)	Determinants of FII in India: An empirical analysis	2004–2011	I	FII inflows, Nifty-50 index, S&P 500, WPI, exchange rate,	ARDL	The study concludes that Indian equity market returns have negative short-run and positive long-run effects on FII inflows, whereas US equity market returns have positive and significant influence on FII flows in the long run but positive and insignificant influence on FII flows in the short run. It can be concluded that FII inflows to India are essentially determined by exchange rate, domestic inflation, domestic equity market returns and returns and risks associated with US equity market.
18	Arora (2016)	The relation between investment of domestic and foreign institutional investors and stock returns in India	2007–2013	I	FII equity flows, domestic equity inflows and BSE sensex	Granger causality and VAR analysis	The study has contrary findings to earlier studies that MF-investments has no effect on future stock returns and DIIs have positive significant relationship with future stock returns. It also adds on the presence of weak evidence of negative relation between FII investment and future stock returns.

Source: Prepared by authors.

Table 2. List of Sample Developed and Developing Countries

Country Name	Code	Country Name	Code
The United States of America	USA	Brazil	BRA
The United Kingdom	UK	Russian Federation	RUS
Australia	AUS	China	CHN
Singapore	SGP	India	IND
Canada	CAN	South Africa	SAF
Switzerland	SWL	Indonesia	IDN
New Zealand	NZL	Thailand	THA
Germany	GER	Malaysia	MYS
France	FRA		
Italy	ITA		
Japan	JPN		

Source: Prepared by authors.

This study is limited to 19 countries having data period of 10 years from year 2004–2013, which has been collected form Bloomberg database. Bloomberg is world-renounced database in accuracy and availability of variables used in study, except for freedom index which is extracted from heritage foundations.[2] Variables have been identified as per extensive literature review tabulated in Table 3.

Choice of Variables

The selected variables enlisted in Table 3 have been captured as per major research studies undertaken across the globe.

Research Methodology

As per the literature review, it was found that OLS, Granger causality, VAR, ARDL, multiple regression, panel OLS, fixed effects and random effects are the methods which have been extensively used in earlier studies. In this study, we have used two research methods, one is static and other is dynamic panel data modelling.

Considering static panel data modelling, the first method employed was pooled OLS estimation. Breusch–Pegan multiplier test rejects the null hypothesis of using pooled OLS estimation over random effects. The next step is to check the applicability of fixed effect or random effect in static model approach. Fixed effect panel are likely to be superior on theoretical grounds, as they control for time variant heterogeneity across the countries and provide relatively robust results. Hausman (1978) test indicates the choice between fixed and random effect specifications.

In general, the models for determinants of FPIs can be studied using following specifications:

$$\begin{aligned} FPI_{it} &= c + \sum_{j=1}^{J} \beta_i X_{it}^j + \sum_{k=1}^{K} \beta_k Y_{it}^k + \sum_{l=1}^{L} \beta_l Z_{it}^l + e_{it} \\ e_{it} &= \upsilon_i + u_{it} \end{aligned} \tag{1}$$

where, X, Y and Z are the different vectors of pull and push determinants. This equation represents the static nature of model. The article adopts several approaches to model the determinants of FPI flows.

Table 3. Description of Sample Variables

Ratio/Variable	Explanation	Formula/Database	Literature Review
1. FPI/GDP	Foreign portfolio investment inflows as % of gross domestic product	(FPI inflow/GDP)*100	Rai and Bhanumurthy (2004), Srinivasan and Kalaivani (2015)
2. GDP growth	Growth in GDP as %	$((GDP_t/GDP_{t-1})-1)*100$	Choudhry and Smiles (2004), Fama (1990), Geske and Roll (1983), Vardhan and Sinha (2012)
3. Imports/reserve	Total imports to reserve	(Imports/reserve)	Dua and Garg (2013)
4. Interest rate differential	Difference of interest rates in home country and LIBOR (USA)	Home country interest rate-LIBOR	Bhasin and Khandelwal (2013), Dua and Garg (2013), Fleming (1962), Mukherjee, Bose and Coondoo (2002), Mundell (1963), Reinhart (2005),
5. Trade openness	Total trade as % of GDP	(Trade/GDP)*100	Alam, Arshad and Rajput (2013), Ang (2008), Aw and Tang (2010), Banga (2003), Dua and Garg (2013), Singhania and Gupta (2011)
6. Freedom index	Weighted average of all freedom indexes including fiscal freedom, government spending, business freedom, labour freedom, trade freedom, investment freedom and financial freedom	Extracted from Heritage foundation[3]	Sambharya and Rasheed (2015)
7. Stock market index	Home country stock market index	Extracted from Bloomberg database	Bohl and Brzeszczyński (2006), Boyer and Zheng (2009), Cohen, Gompers and Vuolteenaho (2002), Froot, O'Connell and Seasholes (2001)
8. US stock market index	US stock market index, S&P-500	Extracted from Bloomberg database	Bhasin and Khandelwal (2013), Srinivasan and Kalaivani (2015)
9. Exchange rate	Exchange rate of home country in terms with US$	Extracted from Bloomberg database	Bhasin and Khandelwal (2013), Dua and Garg (2013), French and Vishwakarma (2013), Rai and Bhanumurthy (2004).

Source: Prepared by authors.
Note: Detailed explanation of variables is enclosed in appendices.

Based on theoretical framework and literature review, variables have been identified. We enlist four models with combinations of different pull and push factors. As specified by portfolio balanced framework, the study tried to determine the significance of variables such as credit worthiness of host country, stock markets return of host country, interest rate differentials and many others.

Model 1. FPI_GDP = f(freedom index, exchange rate, GDP, interest rate differential, reserve to imports, stock market index, trade)

Model 2. FPI_GDP = f(freedom index, exchange rate, GDP, interest rate differential, stock market index, trade)

Model 3. FPI_GDP = f(exchange rate, GDP, interest rate differential, reserve to import, stock market index, trade)

Model 4. FPI_GDP = f(exchange rate, interest rate differential, reserve to import, stock market index, trade, crisis dummy)

The estimated coefficients and their corresponding significant results for static models are shown in Table 4.

Another methodology applied here is dynamic panel data modelling. Majumder and Nag (2014) specified that total capital inflows are more voluminous, more volatile and more persistent than net inflows. FPIs being the part of foreign capital inflows are also considered to be volatile and is affected by its own lagged values (measuring the presence of persistence effect). Therefore, the use of lagged independent variable adds dynamic nature to analysis. Now the model can be prescribed as follows, considering lagged value of dependent variable.

$$\begin{aligned} FPI_{it} &= c + \delta FPI_{i,t-1} + \sum\nolimits_{j=1}^{J} \beta_i X_{it}^j + \sum\nolimits_{k=1}^{K} \beta_k Y_{it}^k + \sum\nolimits_{l=1}^{L} \beta_l Z_{it}^l + e_{it} \\ e_{it} &= \upsilon_i + u_{it} \end{aligned} \tag{2}$$

FPI denotes foreign portfolio investment of country i and time t with i = 1…..N, and $t = i \ldots T$. c is the constant term, X_{it}^j, Y_{it}^j, Z_{it}^j are explanatory variables (vector of pull and push factors). e_{it} is disturbance term with unobserved country-specific effects ν_i and u_{it} idiosyncratic error, where $\upsilon_i \sim IIN(0, \sigma_\upsilon^2)$ and $u_i \sim IIN(0, \sigma_u^2)$. Here, one period lag of dependent variable makes the specification dynamic, and d coefficient denotes speed of adjustment. Equation (3) is taking differenced form to remove the unobserved country-specific effects.

$$FPI_{it} = c + \delta \Delta FPI_{i,t-1} + \sum\nolimits_{j=1}^{J} \beta_i \Delta X_{it}^j + \sum\nolimits_{k=1}^{K} \beta_k \Delta Y_{it}^k + \sum\nolimits_{l=1}^{L} \beta_l \Delta Z_{it}^l + e_{it.} \tag{3}$$

In static panel data modelling, estimation is done using fixed and random effects. Using lagged dependent variable yield, the model dynamic in nature and least square estimation in this case provide bias and inconsistent results (Baltagi, 2008). To overcome this situation, Arellano and Bond (1991) suggested the usage of instrumental variables by including lag of dependent and independent variables. The inclusion of dependent lagged variable as regressor in Equations (2) and (3) makes the model dynamic and the problem of endogeneity arises. To account endogeneity problem, following García-Herrero, Gavilá and Santabárbara (2009) and Athanasoglou, Brissimis and Delis (2008), we use differenced generalized method of moments (GMM) proposed by Arellano and Bond (1991).

Analysis and Findings

Tables 7A and 7B in the Appendices depict descriptive statistics of developed and developing counties. FPI as percentage GDP is quite higher in developing countries than developed countries. In terms of creditworthiness, developed nations are capable to pay off their debts. This is a positive sign for short-run investors, if host country have sufficient reserves in their hand. Developed countries are ahead in interest rate differentials, trade openness and GDP growth, which simply means developing countries are capable enough to mould their macroeconomic policies in favourable to foreign institutional investors.

Table 4. Results Static Model of Developed Countries

	Model 1		Model 2		Model 3		Model 4	
	Dependent Variable as FPI_GDP							
Independent variables	Coefficient	*t*-statistic	Coefficient	*t*-statistic	Coefficient	*t*-statistic	Coefficient	*t*-statistic
Freedom index	–0.564,904	–0.235,894	–0.054,252	–0.024,114				
Exchange rate	0.591,116	0.627,472	0.390,716	0.441,342	0.294,183	0.451,35	0.028,01	0.044,155
GDP growth	0.102,364	0.077,663	0.099,073	1.285,653	0.074,233	1.125,74		
Interest rate differential	–0.365,696*	–1.585,365	–0.345,171*	–1.519,146	–0.108,41	–0.756,24	–0.120,354	–0.842,806
Reserve to import	0.184,899	0.650,846			0.098,386	0.451,767	0.087,842	0.407,88
Stock market	1.775,513***	3.670,697	1.868,307***	4.065,078	1.369,682***	3.837,208	1.648,62***	4.152,507
Trade	–2.381,521**	–2.232,718	–2.223,841***	–2.153	–1.744,79***	–2.544,03	–1.100,207***	–1.986,386
US stock market returns					–1.050,92***	–2.495,56	–1.171,089***	–2.703,167
Crisis dummy							0.147,784	1.488,082
Constant	–0.015,18	–0.299,277	–0.027,833	–0.597,465	0.037,584	0.832,701	–0.075,675	0.939,134
R-square	0.362,49		0.359,069		0.292,19		0.299,865	
F-statistic	3.146,266		3.361,389		5.366,52		5.567,842	
Prob (F-statistic)	0.000,4		0.000,265		0.000,0		0.000,00	
Hausman specification test (Chi-square statistics)	5.77,2		3.859,167		3.062,574		2.763,192	
Prob.	0.572,5		0.569,9		0.801,0		0.877,9	
Applicability of models	Random Effect		Random effects		Random Effect		Random Effect	

Source: Authors' calculations.
Note: ***, **, * significant at 1%, 5% and 10%, respectively.

Tables 8A and 8B in the Appendices showcase the correlation matrix for all listed variables in developed as well as developing countries. Implicitly, the absence of high correlation is observed among the variables. To run regression on data, it must be ensured that data sets should be free from the problems of multicollinearity, hetroscedasticity, autocorrelation and stationarity. By applying few specification tests, we found the eligibility of data sets to run regression-based static and dynamic approach.

To select fixed and random effect approach, we estimate Equation (1) with proposed models as explained earlier and perform Hausman specification test. The null hypothesis of the test signifies the applicability of random effects. Results of each model with Hausman test are marked in Tables 4 and 5 for developed and developing countries, respectively.

In case of developed countries, all models failed to accept the applicability of fixed effect as Hausman test confirms the applicability of random effect model. As per the specifications of the aforesaid econometric models, all four models accept the significant effects of domestic stock market performance and trade on foreign portfolio in investment with addition to interest rate differentials in Models 1 and 2, respectively. Models 3 and 4 demonstrate the significant effect of US stock market performance on capital inflows summarized in Table 4.

In developing countries, all models are having the specifications of fixed effects. The estimation of the model starts with Hausman test, which examine the hypothesis that random effect model is applicable to panel data analysis. Here freedom index, interest rate differential, trade, domestic stock market returns affects the inflows of foreign portfolio investments. Entrant of crisis dummy in Model 4, negatively but significantly affects the foreign portfolio investment. As explained, earlier portfolio investments are like flight investments which comes when economy is in upswing and flies away in case of turmoil. Due to global financial crisis in the year 2008, there is intense decline in inflows of FPI in developing countries. Normally investors lose the confidence of investing in the countries where they had invested previously at a very high rate. After global financial crisis, when economy stabilizes, the portfolio investors regained the confidence in investing in these countries, looking at the potential recovery and institutional factors. The results of static models are outlined in Table 4 for developed countries and Table 5 for developing countries.

This motivates the foreign portfolio investors to participate in such countries where stock market returns are quite high. These results of stock market returns on inflows of foreign portfolio investors are consistence with the findings of Bohn and Tesar (1996), Clark and Berko (1996), Brennan and Cao (1997), Kim and Wei (2000) and Richards (2005). The other variable is trade openness which clearly indicates the openness of economy with the whole world. This is again positive and significant, indicating that trade openness increases the confidence of foreign portfolio investors towards the open policies of the country.

As mentioned earlier, least square estimation leads to inconsistent and bias results taking lagged values of dependent variable as regressor. Therefore, we use GMM to account the problem of endogeneity, heteroskedasticity and consistency. Sargan and Wald test statistics are the test for over-identifying restrictions in the model and goodness of fit, respectively.

Lagged dependent variable of FPI comes out to be highly significant, which confirms the dynamic specifications for the model and justified the usage of GMM methodology. The coefficient of lagged dependent variable as regressor is 0.292, which specifies the moderate degree of persistence of FPI flows, and positive value signifies that countries are able to retain foreign portfolio investment from one year to another. Freedom index, trade openness, GDP growth, interest rate differential, home country stock market and crisis dummy are found to be significant determinants of foreign portfolio investment. GDP growth rate is found to be negatively significant, which supports the argument of arbitrage process. However, the first lag is positively significant to FPI inflows, which specifies the attractiveness to foreign

Table 5. Results of Static Model of Developing Countries

	Model 1		Model 2		Model 3		Model 4	
	Dependent Variable as FPI_GDP							
Independent variables	Coefficient	*t*-statistic	Coefficient	*t*-statistic	Coefficient	*t*-statistic	Coefficient	*t*-statistic
Freedom index	13.898,49*	1.823,765	8.294,055	0.884,105				
Exchange rate	–0.031,982	–0.656,601	–0.043,534	–0.734,822	–0.044,848	–0.780,601	–0.023,776	–0.454,433
GDP growth	–2.124,023	–0.438,395	–0.787,455	–0.172,732	1.756,39	0.388,757		
Interest rate differential	4.277,525***	1.241,179	–5.274,566*	–1.331,575	0.109,133	0.026,775	3.387,727	0.905,483
Reserve to import	32.454,38	0.501,919			105.725*	1.512,014	169.901,2	2.677,386
Stock market	0.003,221***	1.591,965	0.004,097*	1.802,53	0.001,62	0.680,288	0.002,85***	1.315,682
Trade	6.882,696**	5.342,656	5.586,302***	4.528,057	8.096,139***	5.888,473	5.857,588***	4.185,777
US stock market returns					0.192,839***	3.877,487	0.154,979***	3.368,331
Crisis dummy							–99.958,85***	–3.542,067
Constant	–454.639,4	–3.424,504	–731.261,4	–1.296,172	–562.226,1	–3.4753	–457.807,4***	–3.237,639
R-square	0.751,353		0.392,139		0.531,867		0.615,302	
F-statistic	6.730,281		3.275,189		4.000,4		6.512	
Prob (F-statistic)	0.000,0		0.000,0		0.000,0		0.000,0	
Hausman specification test (Chi-square statistics)	40.149,567		20.654,584		35.972,544		21.182,040	
Prob.	0.000,0		0.002,1		0.000,0		0.003,5	
Applicability of models	Fixed Effects		Fixed Effects		Fixed Effects		Fixed Effects	

Source: Authors' calculations.

Note: ***, **, * significant at 1%, 5% and 10%, respectively.

Table 6. Dynamic Panel Estimates for FPI Determinants

Variables	Coefficient	*t* statistics	*p*-value
FPI			
FPI_{t-1}	0.292,097,2**	2.20	0.025
FPI_{t-2}	0.420,266,4*	1.62	0.086
Freedom Index			
$Freedom\ Index_{t}$	47.570,99*	1.91	0.056
$Freedom\ Index_{t-1}$	28.113,04*	1.66	0.096
$Freedom\ Index_{t-2}$	18.550,99**	2.21	0.027
Trade Openness			
$Trade\ Openness_{t}$	2.791,432**	2.29	0.022
$Trade\ Openness_{t-1}$	2.052,825**	2.49	0.013
$Trade\ Openness_{t-2}$	–3.565,473*	–1.85	0.065
GDP Growth			
$GDP\ Growth_{t}$	–0.576,41**	–2.03	0.042
$GDP\ Growth_{t-1}$	1.635,653**	–2.26	0.024
$GDP\ Growth_{t-2}$	0.124,657,2	0.89	0.376
Interest rate $differential_{t}$	–0.333,955,4**	–2.23	0.026
Home country stock $market_{t}$	0.648,973,8*	1.63	0.098
US stock $market_{t}$	0.086,374	0.32	0.746
Crises Dummy	–0.244,261,1**	–2.09	0.036
Constant	–397.989,4	–2.04	0.042
Test Order	Prob. (*p*-value)		
AR(1)	0.709,9		
AR(2)	0.435,5		
Sargan Statistics (*p*-value)	0.836		
Wald Test (Chi-sq)	0.000	Chi-square	2,254.10 (15)

Source: Authors' calculations.

Notes: ***, **, * significant at 1%, 5% and 10%, respectively.
Sargan-statistic: The test for over-identifying restrictions in GMM dynamic model estimation.
AR(1): Arellano–Bond test that average auto covariance in residuals of order 1 is 0 (H0: no autocorrelation).
AR(2): Arellano–Bond test that average auto covariance in residuals of order 2 is 0 (H0: no autocorrelation).
Italics means lag variables.

investors to grab higher profits in light of good economic conditions. Considering post estimation tests namely AR (1) and AR(2), coefficients are found to be insignificant which implies the absence of first and second order autocorrelation in data. This does not yield to inconsistency in results. Inconsistency will imply, if second order autocorrelation is present (Arellano & Bond, 1991). Wald test statistics gives chi-square value as 2,254.10 at 15 degree of freedom, rejecting the null hypothesis that estimated coefficient are jointly significantly different from zero, meaning that model is having predictive power (see Table 6).

Table 7A. Descriptive Statistics of Foreign Portfolio Investment of Developed Countries

	FPI/GDP	IMPORTS/ RESERVE	STOCK_ MARKET	TRADE	FREEDOM_ INDEX	EXCHANGE_ RATE	GDP	INTEREST_ RATE	US_STOCK MARKET
Mean	0.847,348	5.600,033	6,812.962	67.700,56	73.630,91	10.113,8	1.949,35	–0.034,546	1,315.501
Median	0.783,488	5.079635	4,754.3	44.563,38	75.9	1.286,094	2.050,623	–0.17	1,257.62
Maximum	10.035,88	16.57007	41,434	345.424,4	88.9	117.753,5	15.240,38	4.823	1,848.36
Minimum	–6.064,334	0.591798	728.61	18.457,55	57.2	0.499,772	–5.637,953	–4.105	903.25
Std. Dev.	1.821,035	3.774734	7,357.494	76.624,01	9.008,485	28.661,96	2.816,088	1.753,336	238.189
Skewness	0.890,146	0.750912	2.786,229	2.762,525	–0.303,717	2.945,939	0.668,383	0.616,184	0.580,665
Kurtosis	10.360,04	3.179426	11.646,17	9.168,722	1.908,639	9.893,916	8.079,859	3.264,62	3.546,381
Observations	110	110	110	110	110	110	110	110	110

Source: Authors' calculations.

Table 7B. Descriptive Statistics of Foreign Portfolio Investment of Developing Countries

Mean	70.383,25	0.839,809	12,878.79	68.348,02	56.668,75	1,206.115	5.711,730	8.708,807	1,315.501
Median	52.462,14	0.716,151	3,449.370	48.586,56	56.000,00	30.961,46	5.747,571	8.665,000	1,257.62
Maximum	573.125,7	2.203,708	69,304.81	185.807,2	66.300,00	10,461.24	14.162,40	20.41000	1,848.36
Minimum	–172.32	0.254,273	449.960,0	17.693,73	49.800,00	1.672,829	–7.820,885	0.380,000	903.25
Std. Dev.	119.492,8	0.449,596	17,867.82	45.425,89	4.672,099	3,150.948	3.325,056	4.232,966	238.189
Skewness	1.135,305	0.919,107	1.742,949	1.059,472	0.466,072	2.285,779	–0.843,984	0.587,574	0.580,665
Kurtosis	6.026,746	3.123,303	5.182,007	2.866,002	1.854,880	6.263,134	5.863,527	3.610,892	3.546,381
Observations	80	80	80	80	80	80	80	80	80

Source: Authors' calculations.

Table 8A. Correlation Matrix of Listed Variables of Developed Countries

DEVELOPED	FPI/GDP	INTEREST_ RATE	GDP	EXCHANGE_ RATE	FREEDOM_ INDEX	IMPORTS/ RESERVE	TRADE	STOCK_ MARKET	US_STOCK MARKET
FPI/GDP	1.000,000	–0.112,872	0.211,806	0.060,880	0.070,203	–0.083,726	0.173,700	–0.108,52	0.119,261
Intt_rate	–0.112,872	1.000,000	0.190,760	–0.356,707	0.350,700	0.189,523	–0.157,689	0.006,266	–0.009,03
GDP	0.211,806	0.190,760	1.000,000	–0.115,351	0.355,159	–0.083,62	0.511,017	–0.167,13	0.183680
Exchange_rate	0.060,880	–0.356,707	–0.115,35	1.000,000	–0.087,386	–0.411,978	–0.162,021	–0.239,54	0.000821
Freedom_index	0.070,203	0.350,700	0.355,159	–0.087,386	1.000,000	0.143,512	0.416,391	–0.376,14	–0.0164
Imports/reserve	–0.083,726	0.189,523	–0.083,62	–0.411,978	0.143,512	1.000,000	–0.285,221	0.182,587	–0.088716
Trade_	0.173,700	–0.157,689	0.511,017	–0.162,021	0.416,391	–0.285,221	1.000,000	–0.128,27	–0.011048
Stock_market	–0.108,515	0.006,266	–0.167,13	–0.239,535	–0.376,144	0.182,587	–0.128,265	1.000,000	0.084533
US_stock market	0.119,261	–0.009,03	0.183,680	0.000,821	–0.016,4	–0.088,716	–0.011,048	0.084,533	1.000000

Source: Authors' calculations.

Table 8B. Correlation Matrix of Listed Variables of Developing Countries

FPI/GDP	1.000,000	–0.085,094	0.095,020	–0.147,063	0.265,907	–0.153,382	0.238,228	0.224,524	0.111,706
Intt_rate	–0.085,094	1.000,000	–0.055,63	0.137,882	–0.093,47	–0.162,572	–0.015,374	0.412,274	–0.188,555
GDP	0.095,020	–0.055,631	1.000,000	0.205,056	–0.094,696	–0.384,104	–0.100,839	–0.126,4	0.078,456
Exchange_rate	–0.147,063	0.137,882	0.205,056	1.000,000	–0.204,028	–0.299,669	0.280,966	–0.219,44	0.004,459
Freedom_index	0.265,907	–0.093,47	–0.094,7	–0.204,028	1.000,000	–0.584,687	0.256,720	0.235,294	0.042,097
Import/reserves	–0.153,382	–0.162,572	–0.384,1	–0.299,669	–0.584,687	1.000,000	–0.170,842	0.041,99	0.0134,01
Trade	0.238,228	–0.015,374	–0.100,84	–0.280,966	0.256,720	–0.170,842	1.000,000	–0.109,77	–0.023,998
Stock_market	0.224,524	0.412,274	–0.126,4	–0.219,438	0.235,294	–0.041,99	–0.109,77	1.000,000	0.099,554
US_stock_ market	0.111,706	–0.188,555	0.078,456	0.004,459	0.042,097	0.013,401	–0.023,998	0.099,554	1.000,000

Source: Authors' calculations.

Conclusion

The study is related to a panel of 19 countries, 11 developed countries and 8 developing countries, and adopted several approaches to model FII inflows. The first method employed is the pooled OLS estimation, then fixed effect model. And the result of Hausman (1978) tests indicates the applicability of fixed effect or random effect model. Finally, we use GMM suggested by Arellano and Bond (1991) to overcome the issues of endogeneity and unobserved heterogeneity.

The findings of the study are based on Gourinchas and Jeanne (2006) and Herrmann and Kleinert (2014). The last decade had experienced the collapse and surge in global FPI, and this study tried to analyse the driving force towards the vulnerabilities of portfolio investment. The study focuses on the debate whether it is the push or pull factor that determines capital inflows. It is found that both pull and push factors determine the inflow of foreign capital in developed and developing countries. The author observed that foreign portfolio investors have watch on interest rate differentials, policy framework of a country reflected in trade openness, performance of home country stock market and US stock market returns as these variables played significant role. In case of developing countries, it is found that freedom index, trade openness, interest rate differential, stock market performance and US stock market performance had a significant impact over foreign portfolio investment, explicitly in developed countries, interest rate differential, host country stock market performance and US stock market significantly played important role in determining the trends of foreign portfolio investment using static modelling. In the final step of analysis, the study tried to capture the economic relevance of dynamic panel specifications with different determinants associated with pull and push factors in explaining foreign capital inflows considering lagged value. And the conclusion is drawn that foreign portfolio investors are quite volatile as they look for short-term gains, and they are persistent in their investment. Majumder and Nag (2015) highlight the presence of persistence in capital flows. And persistence of capital flows measures the degree of predictability. The results of GMM are consistent with the results of static modelling in terms of determining factors that influence inflow of portfolio investment in host country.

This study is considering very important aspect related to growth and development of economies because capital flows are expected to augment growth of developing countries. Once capital flow moves from rich economies (economies with lower interest rates due to capital abundance) to poor economies (economies with higher interest rates due to capital scarcity), it will escalate the path of economic development in emerging and developing countries. Tong and Wei (2009) suggested that emerging and developing economies should reap out the benefits of foreign capital inflows, when the foreign capital is steady, less volatile in nature and will not disrupt the financial stability of the economy. The present study adds the importance of institutional factors considered in freedom index, along with other variables as important determinants FPI. Freedom index, exchange rate, home country stock market, interest rate differential, US stock market returns, crisis period and many more are analysed and found significant in attracting foreign capital, which in turn need to be captured diligently for policy framework and stabilizing the economy. FPI inflows are having moderate persistency which means its past behaviour affects the current inflows. Policies in past affects the current inflows, and it continues until foreign investors can reap out the benefit out of it. The growing importance of foreign capital flows in emerging and developing countries leads the involvement of government in providing a common platform to raise information flows and better coordination in negotiations and executions of projects (Rasiah, Gammeltoft, & Jiang, 2010).

Managerial and Policy Implications

The pattern of capital inflows in developed and developing economies are different because of dissimilar economic and political structures. From the point of view of host country, especially the developing countries, portfolio flows considered to play pivotal role in bridging the saving investment gap and provide foreign exchange to finance current account deficit. The investors finance in portfolios of different countries to diversify the risk and earn more returns. The market participants may consider this study to predict the future movement of FPI in a country/company based on volatility of stock market returns, exchange rate, interest rate differential, economic environment, trade openness and GDP growth. Various multinational companies are interested in forecasting and managing their portfolios to increase their wealth. They consider that the movement of FPI inflows are predicted by its natural determinants. In other words, portfolio managers are also able to identify the spillover effects of FPIs on various determinants. Hence, they can make out the policies considering attractiveness of firms for FPI by adopting good governance. Regarding the policy framework, policies should be addressed in such a way to attract more capital for long-term perspective rather than for short term. No doubt, foreign capital inflows provide financial stability in host country, so stability in capital inflows should be maintained by attracting more of capital inflows by providing them long-term benefits. Many developing countries such as China and India are having superior track record in attracting FPI inflows. Business friendly environment, strategic policy initiatives for providing economic freedom, and flexible laws can be identified as major forces to attract FPI. Policymakers need to ensure economic & political stability, good law and order mechanism, good governance and also suitable fiscal and monetary policies for higher order capital movement among different countries to attain the objective of economic development.

Limitations

There are inherent limitations in data used in this research. Despite the popularity of economic freedom index and other indices published by Heritage Foundation, these have attracted considerable criticism as well. Scott (1997) argued that this database has missed out some important aspects of regulation. The publisher of the database has adopted causal relationship between economic freedom and economic growth rather than the index itself. However, by using foreign capital inflows, this criticism has been addressed. The study is limited to the period of 10 years. The study is also covering the limitations of statistical tools used and economic environment indicators.

Scope of Further Research

The greater validity of results can be attained using economic freedom with location-specific advantage that a country possesses is likely to influence FPI inflows. However, it is important to replicate the study for additional period of time and additional countries to test whether the relationships uncovered by the present study hold true across the different economic cycles and may provide more viable results. The article has not considered governance index and tax regimes of the countries. These variables can be further studied as the future scope of research.

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Appendices

Definition and List of Variables Undertaken

FPI/GDP: Foreign portfolio investment as the percentage of GDP is considered as the dependent variable to study its trends on variables as specified. In the earlier section, this dependent variable determines its trend 'via' different independent variables as listed.

GDP Growth: Foreign portfolio investors are eager to invest in country with high growth rate linking the interest of higher returns on invested capital. Portfolio investors basically look for short-term gains. They invest when high potential is discovered in country and sell their proportion at the time of distress. GDP growth is a measure to collate the growth of country with the hidden potential identified by the investors.

Imports to Reserve ratio: This ratio is quite popular in literature to study the credit worthiness of countries. The ratio indicated to pay off the debts with reserves. Higher ratio indicates the better liquidity in country which builds the confidence of the investors, if they withdraw their funds, Dua and Garg (2013). It is expected that countries with the sufficient amount of reserve (able to pay off debts) are perceived to be credit worthy and probability of default is quite less.

Interest Rate Differential: Interest rate differentials between the host country and London Interbank Offered Rate also determine the trends of FPI flows. As proposed by capital flows from the country where interest rates are high when other things remain constant. Also consider the importance of interest rates in attracting foreign capital inflows.

Trade Openness: As specified earlier, trade is the measure of policy variable. It reflects how country is open towards business and business activities. It is the measure of country's willingness to adopt the culture of openness to different economies in terms of promoting business activities.

Freedom Index: This index is designed by Heritage Foundation[4] with the weighted average of different index such as supporting new business, presence of corruption, economic policy framework for promoting new ideas and many more. Higher the value of index means higher the facilities of entering in such economies with a lot of favourable policy framework.

Stock Market Index: Domestic stock market returns can influence the foreign portfolio investment in a positive and significant way. Higher stock market returns attract the foreign investors and lower stock market returns repel to invest some more profitable destination.

US Stock Market Returns: Most of the developed and developing economies are highly affected by returns of US stock market. Boom in US stock market affects positively the other depending economies of world, and crises in US stock market hits badly the stock exchange of so many countries. To study this impact on foreign portfolio investment, this variable is incorporated into model.

Exchange Rate: Volatility of exchange rate is expected to have negative impact on inflows of capital in host countries. Higher volatility means higher degree of uncertainty in returns earned by foreign investors. Exchange rate is taken as the effectiveness of macroeconomic policies in host country. Stability in exchange rate is expected to be observed by foreign investors, especially portfolio investors to earn higher returns.

Notes

1. Prepared by UNCTAD and named as Global Quarterly foreign portfolio index for all over the world.
2. One of the famous databases of America related to institutional infrastructure and freedom index.
3. The Heritage Foundation is an American conservative think tank based in Washington, DC.
4. The Heritage Foundation is an American conservative think tank based at Washington, DC.

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